

Technology Stack

Date	4 October 2026
Team ID	6ab2951f1266c775575a6b80
Project Name	Farm Management System
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements. Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	React.js, Redux, Material-UI	For a responsive, component-based UI that handles complex data visualization for farm resources and tracking, ensuring high user productivity.
2	Backend / Server-Side (e.g., Node.js, Python/Django, Java)	Python/Django Rest Framework	Python's data processing and GIS capabilities, combined with Django's robust framework, are ideal for managing multi-source farm data.
3	Database / Data Storage (e.g., MySQL, PostgreSQL)	PostgreSQL with PostGIS	Requires spatial data handling for land mapping and field data, which PostGIS on top of relational PostgreSQL provides efficiently.
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	AWS (RDS, EC2, S3) with Docker	Scalable hosting for growing datasets, Docker for containerization and consistent deployment across environments.
5	Version Control & CI/CD (e.g., GitHub, GitHub Actions)	GitHub with GitHub Actions	Industry standard version control, with seamless integration for automated testing and deployment pipelines to ensure code stability.
6	Third-Party APIs / Other Tools (e.g., Stripe, Twilio, Google Maps)	Google Maps Platform (for GIS/Mapping), AccuWeather API (for climate data), Stripe (for potential payments)	Maps for spatial analysis of farm fields; weather API for predictive analysis and task scheduling; Stripe for any product sale transactions.